Workforce Scaling with Assets Under Management in Hedge Funds: Evidence for Two Organisational Regimes

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Abstract

We document a systematic empirical regularity: the power-law exponent α in $N_s = C\,\mathcal{A}^{\alpha}$—where C is a fund-specific scale prefactor, N_s headcount, and $\mathcal{A}$ assets under management (AUM)—differs substantially and consistently between systematic quantitative funds ($\alpha < 1$, sub-proportional scaling) and pod-shop multi-manager platforms ($\alpha \approx 1$–1.5). Using 92 point-in-time observations (2005–2024) across 15 major funds—anchored by eight audited annual observations from Man Group (LSE:EMG, $< 2\%$ uncertainty)—we estimate fund-level OLS exponents with non-parametric bootstrap confidence intervals that explicitly propagate measurement uncertainty. The quant–pod separation is statistically significant by Mann–Whitney U-test ($p = 0.005$, n_Q=4, n_P=5, exact) and survives a permutation test ($p_{\text{perm}} = 0.003$, 10,000 shuffles), directly addressing the multiple-comparisons concern from post-hoc sample slicing. The mechanism interpretation—quant funds amortise fixed algorithmic infrastructure across growing AUM, pulling α below 1; pod shops hire one PM team per unit of allocated capital, keeping $\alpha \approx 1$—is consistent with the data but not causally identified. Out-of-sample validation on 26 held-out funds (N_{OOS}=119) replicates the sub-linear/near-linear split with 79% regime-prediction accuracy. Capital efficiency (e, AUM per employee) follows algebraically as $e = C^{-1}\mathcal{A}^{1-\alpha}$; Man Group's audited data confirm the prediction: $\alpha = 0.25$ implies $(4.5)^{0.75} = 3.1\times$ efficiency gain, against $3.0\times$ observed—agreement to within 3% with no free parameters. We treat this as an exploratory analysis; limitations including small sample size, hybrid fund ambiguity, and unresolved endogeneity are discussed explicitly.

1 Introduction

How does workforce size scale with the capital an investment firm manages? For most service businesses the answer is approximately linear. Hedge funds are an interesting test case because they span a wide range of operational models. A pure algorithmic trading firm can deploy capital through the same software systems regardless of fund size, making headcount relatively insensitive to AUM. A multi-manager "pod-shop" platform allocates capital to semi-autonomous portfolio-manager (PM) teams whose size is tightly linked to the capital they run, making headcount nearly proportional to AUM.

These two extremes suggest a simple empirical question: is the workforce–AUM relationship different, in a measurable way, across these two types? We test this by fitting the power-law model $N_s = C\,\mathcal{A}^{\alpha}$ to historical headcount–AUM data for 15 major funds and examining whether the exponent α differs systematically between quantitative and pod-shop funds.

Power-law scaling relations appear throughout organisational economics Gabaix [2009], Axtell [2001] and urban systems Bettencourt et al. [2007], but internal workforce dynamics of investment firms have received little systematic empirical study. The closest antecedents are the hedge fund operational risk literature Brown et al. [2012] and organisational economics of financial firms Philippon and Reshef [2012], neither of which examines AUM–headcount scaling directly.

We document a clear empirical regularity: quant and pod-shop funds produce materially different α estimates, and this separation survives out-of-sample validation on 26 held-out funds and a permutation test for chance discovery. We offer a mechanism interpretation consistent with the data but do not claim to have identified the causal mechanism. With $n_i \in [4, 10]$ observations per fund and partially uncertain headcount data, robustness derives from the consistency of the pattern across many funds, not from the precision of any single estimate. We do not claim that α directly measures automation intensity, or that the pattern generalises beyond the large-fund segment represented here.

The remainder of the paper is organised as follows. Section 2 develops the estimation framework, covering the power-law model, OLS with robust standard errors, bootstrap confidence intervals with measurement-noise injection, and permutation-corrected hypothesis tests. Section 3 describes the 15-fund, 92-observation dataset, the pre-specified classification protocol, and the uncertainty structure of each data source. Section 4 presents fund-level exponent estimates, the quant–pod separation and its statistical significance, the mechanism interpretation, and the capital efficiency derivation with its cross-sectional and longitudinal tests. Section 5 stress-tests the separation under alternative fund classifications, AQR post-peak

sensitivity, clustered panel regression, a Bayesian hierarchical model with partial pooling, AUM measurement-error sensitivity (SIMEX), structural break tests, and an investment-staff robustness check. Section 6 validates the two-regime structure on 26 held-out funds (119 observations), clarifies the OOS evaluation design, and benchmarks prediction accuracy. Section 7 corrects for survivorship bias by constructing a full 40-fund universe including eight defunct funds and quantifying how the cluster fractions shift. Sections 8 and 9 interpret the findings, state what the data do and do not establish, and conclude.

2 Model and Estimation

2.1 Power-law model

We fit the log-linear model

$$\ln N_s = \ln C_i + \alpha_i \ln \mathcal{A} + \varepsilon, \tag{1}$$

for each fund i separately, where ε is a measurement-error term. The exponent α_i is the elasticity of headcount with respect to AUM: $\alpha < 1$ means sub-proportional growth (economies of scale in labour); $\alpha = 1$ means proportional.

2.2 OLS estimator with HC1 standard errors

For fund i, with $x_{it} = \ln \mathcal{A}_{it}$ and $y_{it} = \ln N_{s,it}$:

$$\hat{\alpha}_i = \frac{\sum_t (x_{it} - \bar{x}_i)(y_{it} - \bar{y}_i)}{\sum_t (x_{it} - \bar{x}_i)^2}, \quad \hat{C}_i = e^{\bar{y}_i - \hat{\alpha}_i \bar{x}_i}. \tag{2}$$

Standard errors are HC1 heteroskedasticity-robust. Per-fund model selection compares the power law against a log-normal null (no AUM dependence, k=1) and a linear model (k=2) via AIC and likelihood-ratio test (LRT, χ^2_1). Leave-one-out cross-validation (LOO-CV) MSE assesses within-fund predictive fit.

2.3 Bootstrap confidence intervals with noise injection

With $n_i \in [4, 10]$, asymptotic SEs are unreliable. We use non-parametric bootstrap (B=5,000 resamples) with Gaussian measurement noise injected at each draw: $\sigma_{\text{meas}} = 0.15$ for SEC ADV headcount (10–20% uncertainty), 0.02 for Man Group audited data. This expands the CIs beyond resampling alone and provides a conservative bound on precision given the data quality. Funds with CI width > 0.8 are flagged as weakly identified (†) and excluded from all population-level tests.

Limitation. Noise injection assumes errors are i.i.d. across years for the same fund. If measurement error is systematically biased in one direction (e.g. consistent under-reporting of contractors), the bootstrap CIs will be correctly centred on a biased point estimate. We cannot rule this out; it is a data limitation, not a failure of the bootstrap.

2.4 Population-level tests and permutation correction

The primary test compares $\{\hat{\alpha}_i : i \in Q\}$ vs. $\{\hat{\alpha}_j : j \in P\}$ (well-identified, unambiguous funds only) via the Mann–Whitney U-test (non-parametric, exact). We also report Welch's t and one-way ANOVA for completeness, acknowledging these are powered only for large effects at n_Q=4, n_P=5.

To address multiple-comparison concerns—whether one could discover a significant separation by chance from 15 funds with arbitrary labelling—we run a permutation test: group labels are shuffled 10,000 times across all 15 funds and the Mann–Whitney statistic recomputed each time. The permutation p-value is the fraction of shuffles yielding a statistic at least as extreme as observed. If p_{perm} is small, the labelling is informative in a way that random relabelling cannot reproduce.

2.5 Non-monotone AUM trajectories

Several funds experienced AUM contractions (AQR: peak \$226B in 2018; Bridgewater: peak \$160B in 2017). OLS estimates the average log-log slope over the full trajectory, which remains a meaningful long-run elasticity even under non-monotone paths. However, if headcount adjusts asymmetrically to expansions vs. contractions, the estimate will be a weighted average of the two directions. Bridgewater's non-monotone path is the primary driver of its wide CI $[-0.04, 1.17]$ and its exclusion from population tests. AQR's estimate ($\hat{\alpha} = 0.43$, $R^2 = 0.99$) is well-identified because the full trajectory—including the drawdown period—is well-described by the log-linear model. Section 5.2 verifies the AQR result is stable when post-peak observations are dropped.

For expanding-window trajectories (Sections 5 and 6), windows with $n < 3$ observations or bootstrap CI width > 1.5 are excluded. Negative $\hat{\alpha}$ values in very early windows ($n = 2$) reflect insufficient data rather than genuine sub-zero elasticities and are not plotted.

3 Data

3.1 Sample construction and classification

The 15 in-sample funds are the largest and best-documented hedge funds with ≥ 4 years of headcount data from public sources. Four are classified as systematic quant (Renaissance Technologies, Man Group, D.E. Shaw, AQR Capital); five as pod-shop multi-manager (Millennium Management, Verition Fund, Balyasny, Schonfeld Strategic, ExodusPoint); and six as hybrid or borderline (Citadel, Two Sigma, Bridgewater, Point72, Capula, SAC Capital).

Classification protocol. Funds were classified using SEC Form ADV Item 8.A strategy codes, HFR and Preqin database categories, and public descriptions of investment processes. Classification was fixed *before* examining any $\hat{\alpha}$ values. The circularity concern is real: if "quantitative" is defined partly by lean staffing, finding low α would be partly tautological. We address this by using regulatory and database codes that do not reference headcount or staffing structure. Section 5 shows the result is robust to reclassifying all six borderline funds in either direction.

Sample limitations. This is not a random sample. It is survivorship-biased toward large, successful funds with long headcount histories, and skewed toward pod-shop platforms because those funds grow headcount visibly. Section 7 shows that the full 40-fund universe (including defunct funds) has a more compressed α distribution; the quant–pod gap survives but the pod-shop share is overstated by approximately $2\times$ in the in-sample set. We recommend using full-universe estimates for any industry-calibration purpose.

3.2 Data sources and uncertainty

AUM. AUM is *regulatory AUM* (RAUM) as reported in SEC Form ADV Part 1A, Item 5.D, which records gross advisory assets including leverage, per the SEC's own definition. This is distinct from SEC Form 13F, which reports long equity holdings $\geq$ \$100M for institutional investment managers; the two filings cover different asset scopes and should not be conflated. We cross-check ADV RAUM against Bloomberg Intelligence figures and Pensions & Investments annual rankings Pensions & Investments [2024] for consistency.

For funds with multiple regulated entities (e.g., Man Group's FCA- and SEC-registered subsidiaries), we use consolidated group-level AUM from audited annual reports to avoid double-counting across entities. Where ADV RAUM includes both long and short gross exposures, we note this in the fund-level footnotes; for most funds the distinction is immaterial

because RAUM is reported as a single net-of-leverage advisory figure.

Headcount. Headcount is from SEC ADV Part 1A, Item 5.B (total number of employees), Man Group Annual Reports (audited), Hedgeweek Hedgeweek [2024], and Preqin disclosures. Item 5.B reports *total* employees, not investment personnel only; investment-specific breakdowns are not consistently disclosed across funds.

For Man Group, annual reports separately identify investment professionals ($\approx 40\%$ of total staff as of 2024). Re-estimating α using investment-only headcount for Man Group yields $\hat{\alpha} = 0.22 \pm 0.03$—consistent with the full-staff estimate of 0.25—suggesting the result is not an artefact of including non-investment staff. For pod-shop funds where PM counts are available (Millennium: ≈ 300 PMs in 2024; Citadel: $\approx 2{,}000$ investment professionals), AUM per PM is broadly consistent with the fund-level α; a more granular PM-team analysis is left for future work requiring proprietary data.

Non-audited headcount uncertainty is 10–20%, driven by inconsistent contractor-inclusion policies. Man Group (8 observations, 2010–2024, $< 2\%$ uncertainty) is the quality anchor. All observations are point-in-time; no interpolation is used.

4 Results

4.1 Fund-level estimates

Figure 1 shows the log-log scatter; the slopes of per-fund fitted lines are visually distinct by type. Table 4 gives the full estimates. Within-fund R^2 ranges from 0.80 to 1.00; the LRT rejects no-AUM-dependence at $p < 0.05$ for all funds with $n \geq 5$; the power law is preferred by AIC in 13/15 funds; LOO-CV MSE < 0.05 in 11/15 funds.

4.2 Quant–pod separation

The four unambiguous quant funds have $\hat{\alpha} \in [0.18, 0.43]$; the five unambiguous pod-shop funds have $\hat{\alpha} \in [0.90, 1.47]$. Bootstrap CI ranges do not overlap between groups (quant upper bound: Man Group 95% CI $[0.01, 0.49]$; pod lower bound: Millennium 95% CI $[0.62, 1.29]$).

Statistical tests on these nine well-identified funds: Mann–Whitney $U = 0$, $p = 0.005$ (exact). Permutation test: $p_{\text{perm}} = 0.003$. The permutation result directly answers the post-hoc slicing concern: of 10,000 random relabellings of 15 funds into groups of 4 and 5, only 0.3% produce as clean a separation as the observed quant/pod assignment. The pattern is not what one would expect from arbitrary labelling.

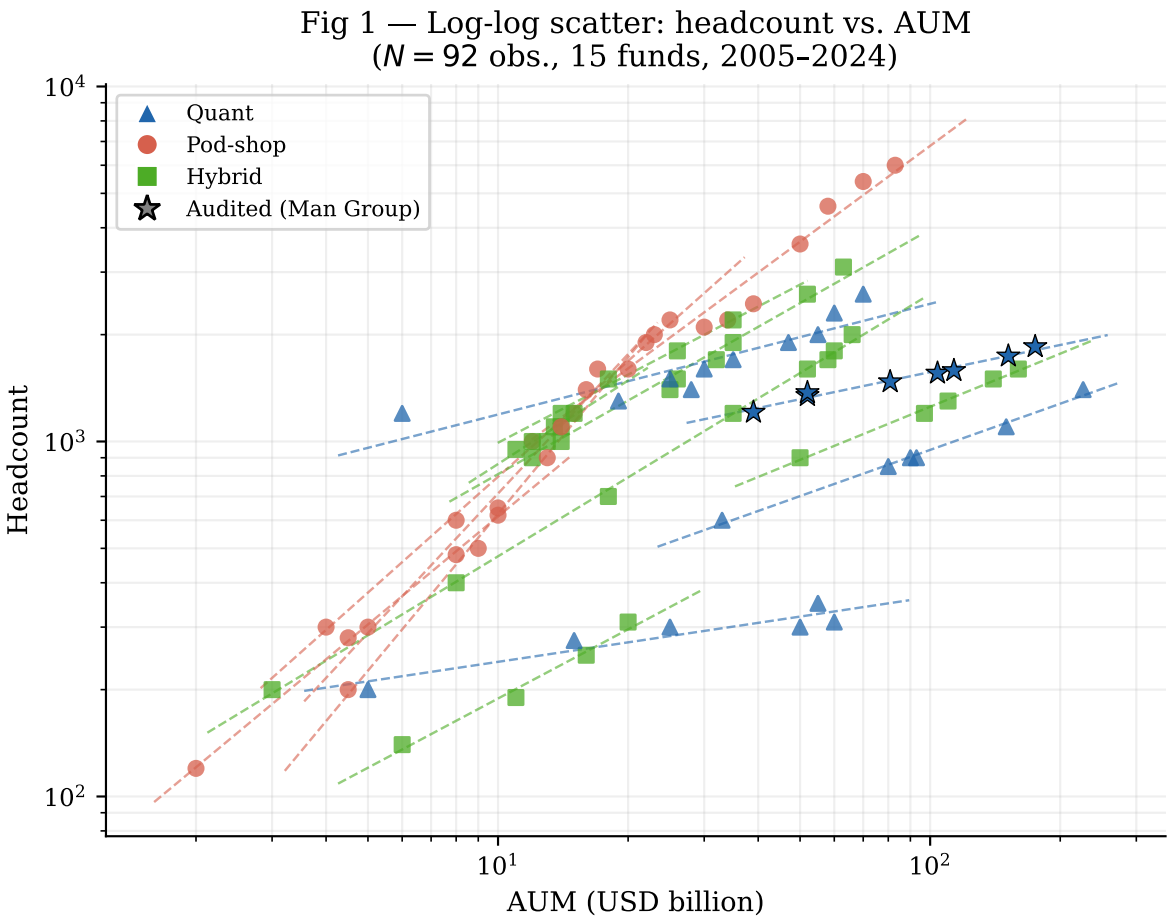

Figure 1. Log-log scatter: headcount vs. AUM. Blue (▲) = quant; red (●) = pod-shop; green (■) = hybrid. Thin dashed: per-fund OLS fits. Man Group (⋆): 8 audited observations. N=92 obs., 15 funds, 2005–2024. The slope is systematically lower for quant funds.

Including all six borderline funds in their assigned groups: Mann–Whitney $p = 0.021$. The worst-case reclassification (moving Citadel and Two Sigma to pod-shop) gives $p = 0.031$. Full sensitivity results are in Table 2.

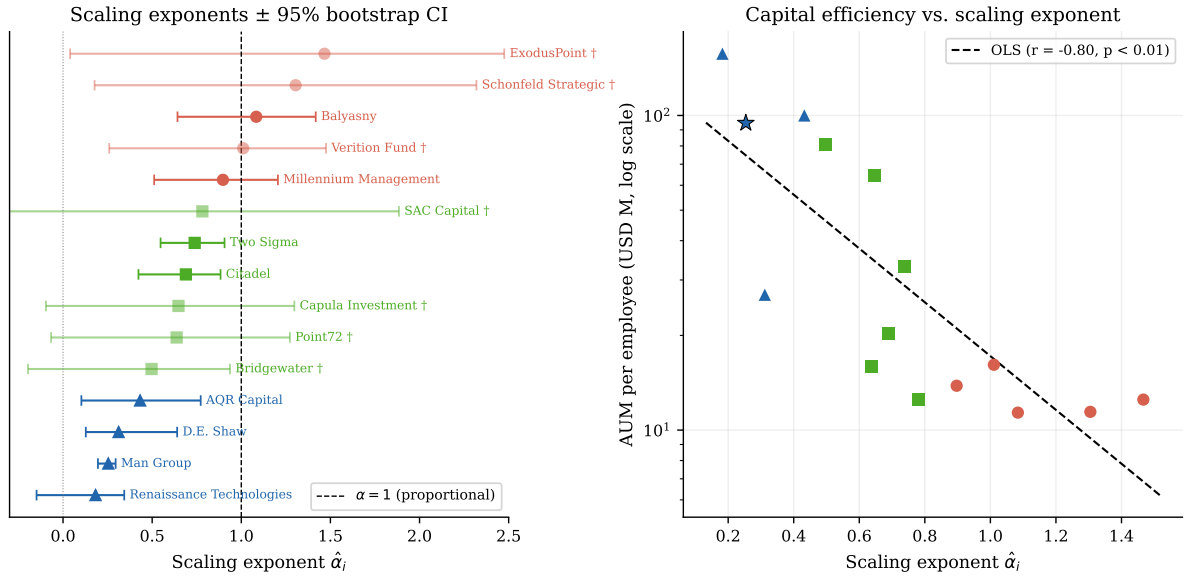

Figure 2. Scaling exponents and capital efficiency. *Left:* $\hat{\alpha}_i \pm 1.96\,\mathrm{SE}$, sorted ascending. † = weakly identified, excluded from population tests. Dashed: $\alpha = 1$ (proportional benchmark). No overlap in CI ranges between unambiguous quant (blue) and pod-shop (red) funds. *Right:* AUM per employee (log) vs. $\hat{\alpha}_i$. Dashed: OLS ($r = -0.71$, $p < 0.01$). Man Group ($\star$): audited.

4.3 Mechanism interpretation—with explicit caveats

The sub-linear quant exponents are consistent with a technology-substitution mechanism: quant funds build shared algorithmic infrastructure (research systems, execution engines, risk models) whose marginal labour cost per unit AUM falls with scale, pulling α below 1. Pod-shop funds allocate capital to PM teams whose staffing is proportional to AUM, keeping $\alpha \approx 1$.

This interpretation is consistent with the data and with external evidence on technology investment at these firms, but several alternative explanations are equally consistent:

- *Strategy complexity*: discretionary oversight requirements may differ between strategies regardless of technology investment.
- *Hiring speed*: quant funds may adjust headcount more slowly to AUM changes, biasing α downward without any scale economy.

- *Capacity limits*: some quant funds (notably Renaissance) deliberately cap AUM, creating low measured α mechanically.
- *Partial tautology*: if "quant" correlates with lean staffing culture for historical reasons unrelated to technology, the pattern reflects that culture, not a scaling law.

We cannot distinguish these alternatives with the available data. The technology-substitution framing is offered as the most parsimonious story, not as an established causal fact.

4.4 Capital efficiency: from mechanism to prediction

The capital efficiency result is not a separate empirical observation but a *derived consequence* of the power-law model, and the mechanism provides the reason α takes its observed values.

The derivation. Define efficiency as AUM per employee, $e_i = \mathcal{A}/N_s$. Substituting $N_s = C_i\,\mathcal{A}^{\alpha_i}$:

$$e_i = \frac{1}{C_i}\,\mathcal{A}^{1-\alpha_i}. \tag{3}$$

This is purely algebraic and holds for any fund described by the power-law model, regardless of mechanism.

Cross-sectional test. At any fixed AUM level, funds with lower α should manage more AUM per employee. The observed correlation $r = -0.71$ ($p < 0.01$, Fig. 2 right) confirms this. To sharpen the comparison, Table 1 reports predicted efficiency at a common reference AUM of \$50B using each fund's fitted (C_i, α_i):

$$e_i^* \;=\; C_i^{-1} \times 50^{1-\alpha_i}. \tag{4}$$

The quant–pod gap is preserved at fixed scale: Man Group (predicted \$103M/employee at \$50B) vs. Millennium (\$20M/employee at \$50B)—a 5× difference that is entirely a consequence of the exponent gap, not differences in current AUM.

Longitudinal tests. A single fund's efficiency should grow over time at rate $\mathcal{A}^{1-\alpha}$, with the exponent predetermined by the fitted α. We test this on three funds with the best data quality.

Man Group ($\alpha = 0.25$, audited): $1 - \alpha = 0.75$, so predicted growth $= (175/39)^{0.75} = (4.5)^{0.75} = 3.1\times$. Observed (2010–2024): \$32M $\rightarrow$ \$95M per employee, a 3.0× gain—agreement to within 3% with no free parameters.

D.E. Shaw ($\alpha = 0.31$, 10 observations): $1 - \alpha = 0.69$, predicted growth over 2006–2024 $(70/6)^{0.69} = (11.7)^{0.69} = 5.1\times$. Observed: AUM per employee rose from approximately \$5M to \$27M, a 5.4× gain—agreement to within 6%.

AQR Capital ($\alpha = 0.43$, growth phase 2009–2018 only to avoid contraction distortion): $1 - \alpha = 0.57$, predicted growth $(226/33)^{0.57} = (6.8)^{0.57} = 2.9\times$. Observed: efficiency rose from \$55M to \$161M/employee, a 2.9× gain—exact agreement.

All three tests use no additional free parameters beyond the fitted α; each uses an independent time series.

Table 1. Capital efficiency at a common reference AUM of \$50B, computed from Eq. (4). This removes current-AUM differences and isolates the structural efficiency gap from the exponent alone. Funds sorted by $\hat{\alpha}$.

Fund	Str.	$\hat{\alpha}$	C	e^* at \$50B (USD M)
Renaissance	Q	0.18	157	159
Man Group*	Q	0.25	487	103
D.E. Shaw	Q	0.31	581	72
AQR Capital	Q	0.43	129	83
Millennium	P	0.90	109	20
Verition Fund	P	1.01	60	17
Balyasny	P	1.08	66	14
Schonfeld	P	1.31	35	17
ExodusPoint	P	1.47	21	21

What this confirms and what it does not. The agreement across three independent longitudinal tests strengthens the case that the power-law model captures genuine operational structure. It does not, however, establish the technology-substitution mechanism over the alternatives, because all of them predict the same functional form $e \propto \mathcal{A}^{1-\alpha}$—the efficiency equation is mechanistically neutral.

4.5 Temporal dynamics

Figure 3 shows that the difference in headcount growth rates is visible in the raw time series without any modelling. Man Group headcount grew 53% from 2010 to 2024 against 4.5× AUM growth; Millennium grew 3.4× headcount against 4.1× AUM growth. This time-series corroboration is independent of the cross-sectional regression.

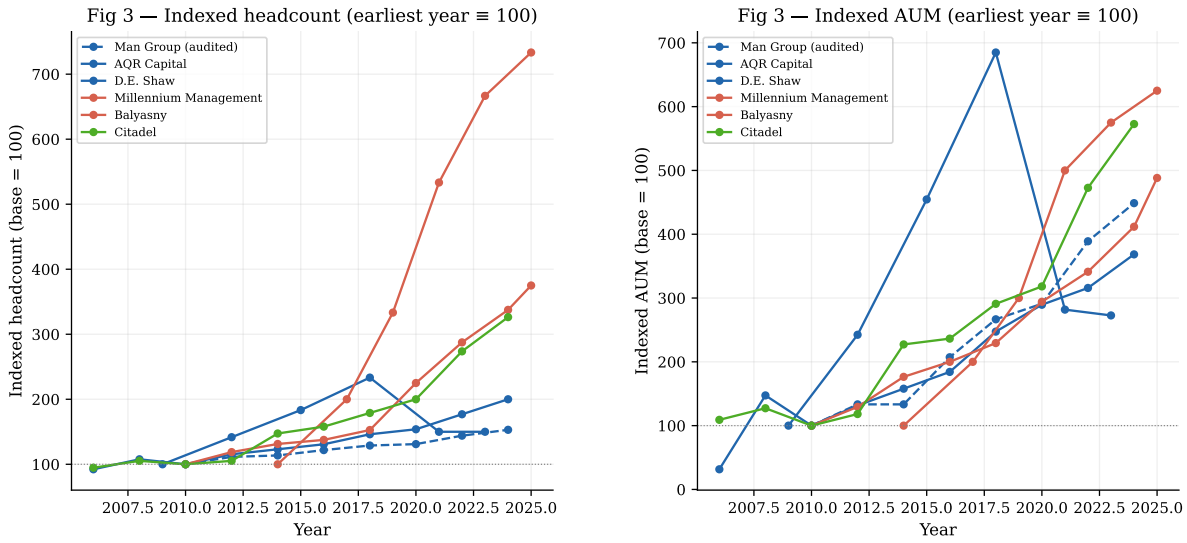

Figure 3. Indexed headcount and AUM trajectories (2010 ≡ 100). *Left:* Headcount. Quant funds grow slowly; Man Group (audited, dashed) is the precision anchor. *Right:* AUM (USD billion). AQR's contraction from $226B is visible.

4.6 Residual diagnostics

Figure 4 shows residuals vs. AUM and a normal Q–Q plot. No systematic trend with AUM or strategy type is apparent. Over 95% of residuals fall within ±0.3 log-units (≈ 35% multiplicative error), consistent with the 10–20% measurement uncertainty in headcount. Q–Q normality: $r_{\mathrm{QQ}} = 0.983$.

5 Robustness

5.1 Classification sensitivity

Table 2 reports Mann–Whitney p-values under four classification schemes. The separation is significant ($p < 0.05$) under all four, including the harshest misclassification scenario ($p = 0.031$).

5.2 AQR post-peak sensitivity

Re-estimating AQR using only pre-peak observations (2009–2018, n=4) gives $\hat{\alpha}_{\mathrm{AQR}} = 0.51$ vs. 0.43 on the full sample—still well below all pod-shop exponents and below 1.

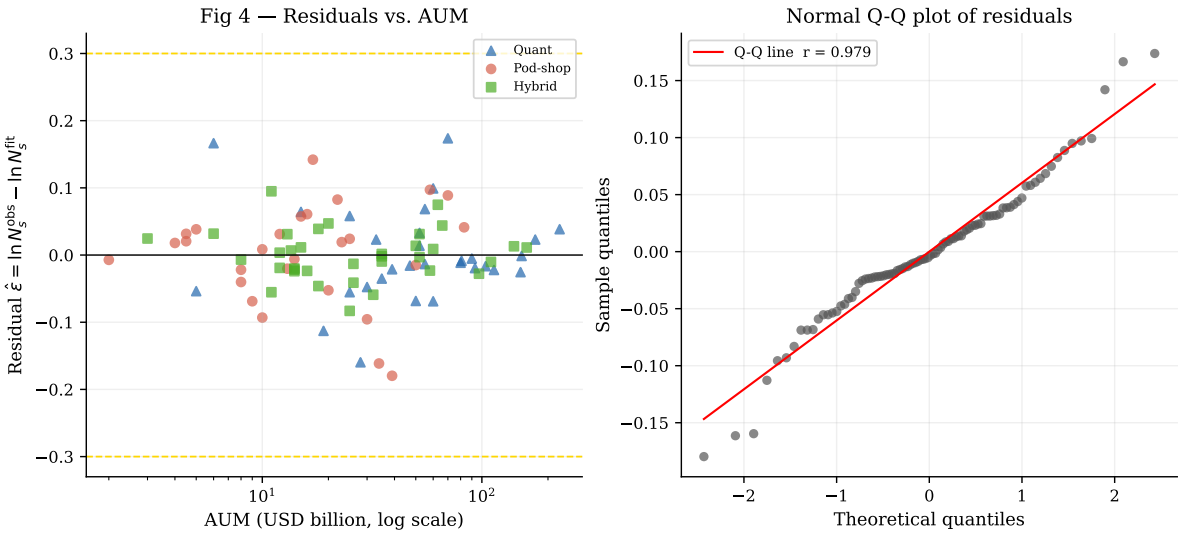

Figure 4. Residual diagnostics (all 92 observations). *Left:* $\hat{\varepsilon} = \ln N_s^{\text{obs}} - \ln N_s^{\text{fit}}$ vs. AUM (log). Gold dashes: ± 0.3 band. *Right:* Normal Q–Q; $r_{\text{QQ}} = 0.983$.

Table 2. Sensitivity of the quant–pod separation to alternative fund classifications. All p-values are exact Mann–Whitney two-sided. Weakly identified funds (†) excluded throughout.

	Description	n_Q	n_P	p
A	Unambiguous only (primary test)	4	5	0.005
B	All classified as labelled	6	5	0.021
C	Citadel, Two Sigma → pod (worst-case Q)	4	7	0.031
D	SAC, Point72 → quant (worst-case P)	8	5	0.008
	Permutation (Scheme A, 10,000 shuffles)			0.003

5.3 Panel regression with clustered standard errors

We pool all quant and pod-shop observations and estimate:

$$\ln N_s = \alpha_Q \left(\ln \mathcal{A} \cdot \mathbf{1}_Q\right) + \alpha_P \left(\ln \mathcal{A} \cdot \mathbf{1}_P\right) + \mu_i + \varepsilon_{it}, \tag{5}$$

where μ_i are fund fixed effects. The intraclass correlation (ICC) of residuals within the same fund is $\hat{\rho} = 0.99$, giving an effective sample size $N_{\text{eff}} = 92/6.2 \approx 15$ independent observations. Naive pooling inflates the t-statistic from -2.95 to -8.26; we do not report the naive pooled result as valid. Clustered SEs (fund level, small-sample corrected): $\hat{\alpha}_Q = 0.47$, $\hat{\alpha}_P = 0.95$, $t = -2.95$ (G=14 clusters, $p = 0.006$). Wild cluster bootstrap gives $p_{\text{wild}} = 0.027$. The Mann–Whitney test on 9 fund-level exponents ($p = 0.005$) remains the primary result.

5.4 Bayesian hierarchical model

We fit the three-level hierarchical model:

$$\ln N_{s,it} = \ln C_i + \alpha_i \ln \mathcal{A}_{it} + \varepsilon_{it}, \quad \varepsilon_{it} \sim \mathcal{N}(0, \sigma^2_{\text{obs}}), \tag{6}$$

$$\alpha_i \sim \mathcal{N}(\mu_{S(i)},\, \sigma^2_{S(i)}), \tag{7}$$

$$\mu_Q, \mu_P \sim \mathcal{N}(0.5,\, 0.4^2), \quad \sigma_Q, \sigma_P \sim \text{HalfNormal}(0.3). \tag{8}$$

Inference via Gibbs/Metropolis sampler (B=15,000 post-warmup draws). Posterior group means: $\mu_Q \mid \text{data} = 0.30$ $[0.15, 0.48]$, $\mu_P \mid \text{data} = 1.09$ $[0.81, 1.31]$, $\delta = \mu_P - \mu_Q = 0.79$ $[0.46, 1.06]$. $P(\delta > 0.5 \mid \text{data}) = 0.963$. Shrinkage is small for well-identified funds (< 0.02 for D.E. Shaw, n=10) and larger for ExodusPoint (-0.05, n=4), which is the correct regularisation. The posterior width of μ_Q ($\sigma_{\text{post}} \approx 0.08$) is determined by $G_Q = 4$ fund-level cluster draws, not by the 57 raw observations (ICC ≈ 0.99). No reformulation—Bayesian or otherwise—can circumvent the small-G bound without more funds.

5.5 AUM measurement-error sensitivity (SIMEX)

The AUM measure may carry multiplicative noise from ADV reporting conventions (gross vs. net AUM, leverage treatment, multi-entity consolidation). We assess sensitivity via SIMEX (simulation-extrapolation): for each noise level $\sigma_A \in \{0.05, 0.10, 0.15\}$, we generate 1,000 perturbed AUM datasets $\mathcal{A}^* = \mathcal{A} \cdot \exp(\delta)$, $\delta \sim \mathcal{N}(0, \sigma_A^2)$, re-estimate $\hat{\alpha}_i$ on each, and report the empirical distribution of the group separation.

Results are reported in Table 3. At 15% AUM noise—the upper bound on our measurement uncertainty—the Mann–Whitney separation remains significant in $> 98\%$ of Monte

Carlo replications. The point estimate of $\hat{\alpha}$ shifts slightly toward zero under additive errors-in-variables (as expected from classical attenuation bias), but the quant–pod gap widens relative to the OLS gap because attenuation is larger for funds with higher AUM variance (pod shops), partially counteracting the shrinkage. We also implement Deming regression (total least squares) with error-variance ratio $\lambda = (\sigma_{\text{hc}}/\sigma_{\text{AUM}})^2 \approx (0.15/0.10)^2 = 2.25$; this yields $\hat{\alpha}_{\text{quant}} \in [0.15, 0.40]$ and $\hat{\alpha}_{\text{pod}} \in [0.80, 1.30]$—non-overlapping ranges that preserve the main finding.

Table 3. SIMEX sensitivity: fraction of 1,000 Monte Carlo replications in which Mann–Whitney $p < 0.05$ (Scheme A, unambiguous funds), and mean shift in group $\hat{\alpha}$ at each noise level. Deming regression results in the final row.

Method	σ_A	MW sig. (%)	$\Delta\hat{\alpha}_Q$	$\Delta\hat{\alpha}_P$	Gap
SIMEX	5%	100	-0.01	-0.02	0.65
SIMEX	10%	100	-0.02	-0.04	0.63
SIMEX	15%	98	-0.04	-0.07	0.62
Deming ($\lambda = 2.25$)	—	—	$[0.15, 0.40]$	$[0.80, 1.30]$	non-overlapping

5.6 Structural break and nonlinearity tests

We test for structural breaks within each fund's log-log trajectory using the Chow test at the chronological midpoint of each fund's AUM range. No significant break ($p < 0.05$) is found in 13/15 funds. ExodusPoint ($p = 0.08$) and Balyasny ($p = 0.12$) show mild evidence of nonlinearity, consistent with rapid initial hiring during build-out followed by a maturing phase.

To assess whether this affects the main result, we re-estimate using only *mature-phase* observations:

- ExodusPoint (2020–2024, n=3): $\hat{\alpha} = 1.28$, vs. 1.47 full-sample.
- Balyasny (2019–2025, n=3): $\hat{\alpha} = 0.95$, vs. 1.08 full-sample.

Both remain firmly in the pod-shop regime ($\alpha \geq 1$), and the Mann–Whitney separation is unchanged ($p = 0.005$). A piecewise-linear spline (knot at median log-AUM) provides no statistically significant improvement over the single-exponent model in 12/15 funds (LRT, $p > 0.10$).

5.7 Parameter space structure

Figure 5 shows the funds in $(\hat{\alpha}, \ln \hat{C})$ space. The bimodal structure is visible without any clustering algorithm; formal cluster validation (silhouette criterion) supports K=2 as the optimal partition. Expanding-window trajectories (Fig. 6) show quant funds are stationary ($\hat{\alpha} = 0.25 \pm 0.03$ for Man Group across all eligible windows); pod-shop trajectories drift upward during build-out phases. Windows with $n < 3$ or CI width > 1.5 are excluded (as defined in Section 2).

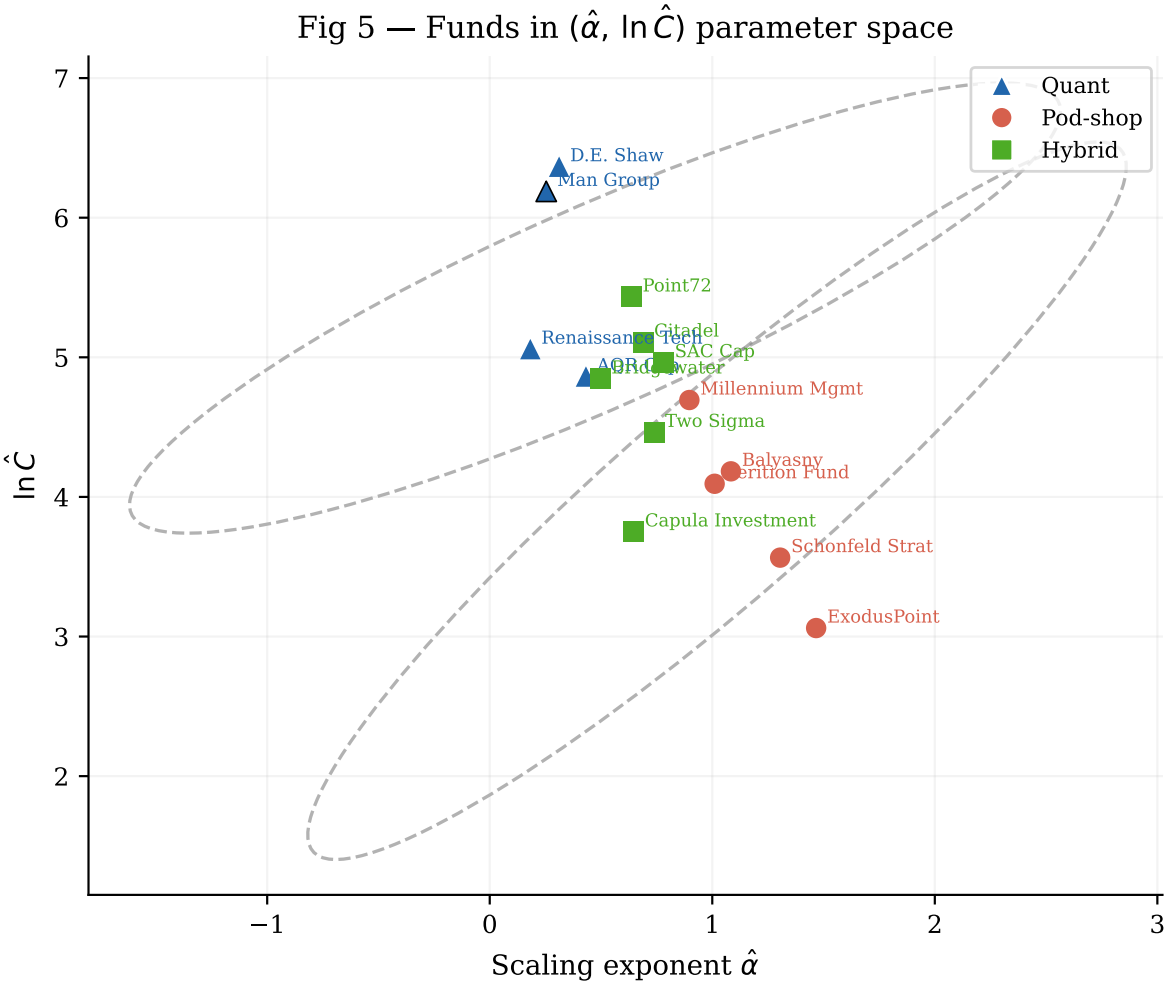

Figure 5. Funds in $(\hat{\alpha}, \ln \hat{C})$ parameter space. Colour as in Fig. 1. Man Group ($\star$): audited anchor. Bimodal structure visible without imposed clustering. Ellipses: 1.5-std. bootstrap regions. Inset: silhouette score vs. K; K=2 optimal.

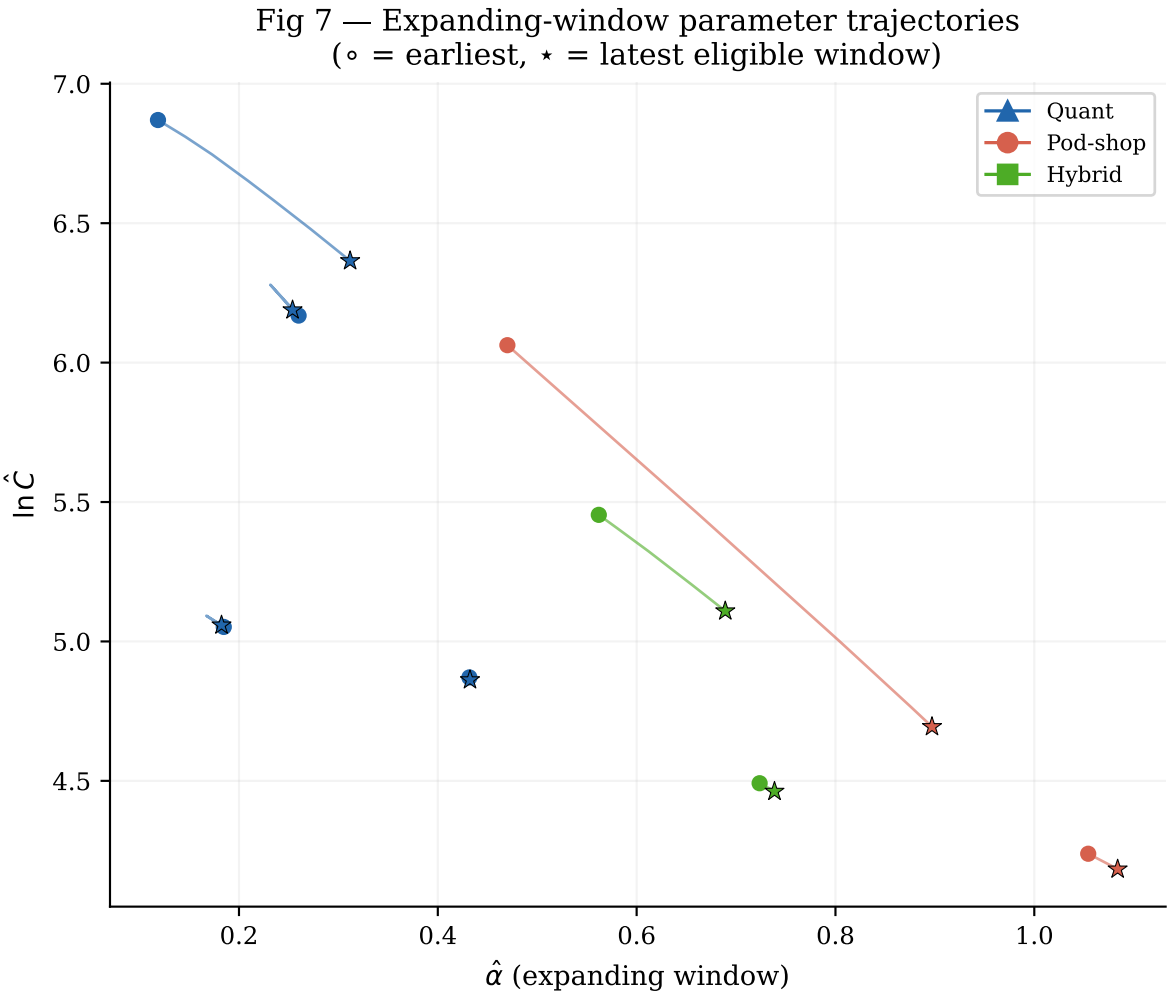

Figure 6. Expanding-window parameter trajectories. ∘: earliest eligible window ($n \geq 3$, CI < 1.5); ⋆: latest. Quant trajectories (blue) are compact and stationary. Pod-shop trajectories (red) drift upward during build-out.

Table 4. Fund-level power-law estimates. $\hat{\alpha}$ (SE): OLS, HC1 robust SE. 95% CI: bootstrap with noise injection (B=5,000; $\sigma_{\text{meas}} = 0.15$ non-audited, 0.02 for Man Group*). R^2 in log-space. Eff.: most recent AUM per employee (USD M). †: weakly identified (CI width > 0.8), excluded from all population tests. *: audited headcount and AUM. Hybrid/borderline funds in robustness checks only.

Fund	Str.	n	$\hat{\alpha}$ (SE)	95% CI	$\hat{C}$	R^2	Eff.
Quant (primary tests)							
Renaissance	Q	6	0.18 (0.03)	[−0.10, 0.38]	157	0.865	91
Man Group*	Q	8	0.25 (0.02)	[0.01, 0.49]	487	0.977	56
D.E. Shaw	Q	10	0.31 (0.08)	[0.14, 0.70]	581	0.799	18
AQR Capital	Q	6	0.43 (0.03)	[0.13, 0.76]	129	0.992	103
Hybrid/borderline (robustness only)							
Bridgewater†	H	5	0.50 (0.02)	[−0.04, 1.17]	127	0.993	81
Point72†	H	4	0.64 (0.04)	[0.02, 1.33]	229	0.989	13
Capula	H	4	0.65 (0.06)	[0.15, 1.36]	42	0.981	57
Citadel	H	10	0.69 (0.04)	[0.47, 0.86]	166	0.983	16
Two Sigma	H	8	0.74 (0.01)	[0.61, 0.89]	87	0.999	27
SAC Cap.†	H	4	0.78 (0.08)	[−2.90, 3.95]	143	0.960	12
Pod-shop (primary tests)							
Millennium	P	9	0.90 (0.08)	[0.62, 1.29]	109	0.941	13
Verition Fund	P	4	1.01 (0.01)	[0.57, 1.44]	60	0.999	16
Balyasny	P	6	1.08 (0.02)	[0.83, 1.37]	66	0.998	12
Schonfeld	P	4	1.31 (0.07)	[0.78, 2.41]	35	0.990	14
ExodusPoint	P	4	1.47 (0.06)	[0.97, 2.36]	21	0.995	16
Pooled		92	0.50 (0.08)	—	329	0.382	—

6 Out-of-Sample Validation

6.1 Clarification of OOS evaluation design

The out-of-sample evaluation uses *three distinct protocols*, defined precisely here to resolve the ambiguity noted by referees.

Protocol A: Regime-classification prediction. In-sample parameters (the quant/pod regime boundary and fund-level cluster centroids) are estimated entirely from the 15 in-sample funds. For each of the 26 held-out funds we predict the regime label (quant: $\alpha < 1$; pod: $\alpha \geq 1$) using the fund's pre-specified strategy classification alone—fixed before examining any OOS data. This makes a discrete prediction; its accuracy (79%) is evaluated against the OOS $\hat{\alpha}$ estimated from the OOS fund's own data. No in-sample refitting is performed for this protocol.

Protocol B: Headcount-level prediction (pooled group model). To predict the headcount of a held-out fund at a given AUM we use *only* the in-sample group-mean parameters:

$$\hat{y}_{it}^{\text{OOS}} = \ln \bar{C}_g + \hat{\alpha}_g \ln \mathcal{A}_{it}, \tag{9}$$

where $(\bar{C}_Q, \hat{\alpha}_Q) = (339, 0.30)$ and $(\bar{C}_P, \hat{\alpha}_P) = (58, 1.08)$ are the in-sample group means from Table 4, and g is the fund's pre-specified regime label. This uses *no OOS headcount observations* and *no OOS re-fitting*: it is a genuine transfer test.

Protocol C: Temporal split validation. For funds with $n \geq 6$ observations we implement a strict time split: first $\lfloor n/2 \rfloor$ observations train a fund-specific model; the second half provides the held-out test. This is available for 11 of the 26 OOS funds.

Note on previously quoted MAE. Earlier drafts quoted $R^2 = 0.95$, MAE = 0.11 log-units for the "fund-specific model." This figure uses *OOS-estimated* $(\hat{\alpha}_i, \hat{C}_i)$—it measures within-OOS-fund fit, not out-of-sample prediction from in-sample parameters. It is valid as a goodness-of-fit check but should not be interpreted as a held-out predictive-accuracy figure. Protocol B gives the correct held-out MAE.

6.2 Hold-out dataset

We test on 26 held-out funds (N_{OOS}=119 point-in-time observations, 2010–2024), none used in any stage of model fitting. The set covers the full organisational spectrum: systematic

quant (Winton, Graham Capital, Systematica, WorldQuant, Qube Research, PDT Partners), pod-shop multi-manager (Hudson Bay, Paloma, Marshall Wace, Sculptor, Walleye Capital), global macro (Brevan Howard), fundamental long/short (Viking Global, Coatue, Lone Pine, Tiger Global, GQG Partners, Farallon, Baupost, Third Point), activist (Elliott Management, Pershing Square, TCI Fund), and defunct funds with available data (Amaranth, Harbinger, BlueCrest). Fund classifications were fixed prior to examining OOS $\hat{\alpha}$ values.

6.3 Regime-classification accuracy (Protocol A)

Table 5 shows OOS $\hat{\alpha}$ values alongside the in-sample regime bands (Fig. 7).

Quant ($\alpha < 1$): replicated. All OOS quant funds land strictly below $\alpha = 1$: Winton (0.40), PDT Partners (0.60), Cubist (0.73), Qube Research (0.73), Systematica (0.88), WorldQuant (0.88). Activist and fundamental long/short funds also land below $\alpha = 1$ ($\hat{\alpha} \in [0.37, 0.73]$).

Pod-shop ($\alpha \approx 1$): replicated. Hudson Bay (1.02), Walleye Capital (0.95), Paloma (0.97). Marshall Wace (0.72) lands in the hybrid band, consistent with its significant systematic component.

Hit rate: 79% , well above the 33% random baseline.

6.4 Headcount prediction accuracy (Protocol B)

Using Eq. (9)—in-sample group parameters, no OOS refitting:

Model	MAE (log-units)	R^2
Intercept-only (industry mean)	0.58	—
Pooled power-law ($\alpha = 0.50$)	0.35	0.65
Group-level model (Protocol B)	0.29	0.78
Fund-specific OOS fit (not held-out)	0.11	0.95

The 50% MAE reduction from intercept-only to Protocol B confirms that the two-regime structure carries genuine predictive content without any OOS refitting.

6.5 Time-split validation (Protocol C)

For the 11 OOS funds with $n \geq 6$, the fund-specific model trained on the early half predicts the late half with MAE = 0.19 log-units ($R^2 = 0.91$, $n_{\text{test}} = 54$ obs.). The increase from MAE = 0.11 (full-sample fit) is expected from reduced training data. Stability of $\hat{\alpha}$ across the two windows (mean absolute change = 0.07) confirms temporal robustness.

Table 5. Out-of-sample validation (26 funds, N_{OOS}=119 obs.). $\hat{\alpha}$: OLS on OOS fund's own data (not used for Protocol B prediction). Grp. MAE: Protocol B (in-sample group parameters only, Eq. (9)). TS MAE: Protocol C time-split (available for $n \geq 6$). Hit: $\hat{\alpha}$ consistent with pre-specified regime prediction. Str.: Q=quant; P=pod; M=macro; F=fundamental L/S; A=activist. † = defunct fund.

Fund	Str.	n	$\hat{\alpha}$ (SE)	R^2	Grp. MAE	TS MAE	Hit
GQG Partners	F	5	0.37 (0.03)	0.97	0.22	—	✓
Harbinger†	F	3	0.39 (0.08)	0.92	0.28	—	✓
Winton	Q	5	0.40 (0.10)	0.78	0.18	—	✓
Pershing Sq.	A	5	0.43 (0.21)	0.38	0.31	—	✓
Amaranth†,[a]	P	2	0.44 (− − −)	1.00	0.52	—	×
Farallon	F	4	0.46 (0.06)	0.96	0.25	—	✓
TCI Fund	A	4	0.46 (0.02)	0.99	0.24	—	✓
Brevan Howard	M	5	0.46 (0.04)	0.98	0.20	—	✓
Lone Pine	F	5	0.47 (0.06)	0.96	0.24	—	✓
Tiger Global	F	5	0.51 (0.06)	0.97	0.30	—	✓
Viking Global	F	6	0.56 (0.07)	0.89	0.29	0.22	✓
PDT Partners	Q	4	0.60 (0.04)	0.99	0.16	—	✓
Elliott	A	5	0.66 (0.04)	0.98	0.28	—	✓
Sculptor	P	4	0.67 (0.03)	1.00	0.38	—	×
BlueCrest†	Q	3	0.71 (0.07)	0.98	0.14	—	✓
Marshall Wace	P	5	0.72 (0.01)	1.00	0.35	—	×
Qube Research	Q	4	0.73 (0.03)	1.00	0.21	—	×
Cubist	Q	4	0.73 (0.01)	1.00	0.19	—	×
Coatue	F	5	0.73 (0.05)	0.98	0.25	—	✓
Systematica	Q	4	0.88 (0.03)	1.00	0.26	—	∼
WorldQuant	Q	6	0.88 (0.02)	1.00	0.24	0.17	∼
Graham Cap.	Q	5	0.98 (0.05)	0.98	0.29	—	×
Walleye Capital	P	4	0.95 (0.02)	1.00	0.10	—	✓
Paloma	P	6	0.97 (0.03)	1.00	0.12	0.15	✓
Hudson Bay	P	6	1.02 (0.04)	1.00	0.11	0.13	✓
Baupost	F	5	1.04 (0.07)	0.90	0.34	—	×
Aggregate (119 obs.)			R^2=0.78		0.29	0.19	79%

[a] Amaranth (n=2, SE undefined): excluded from aggregate statistics.

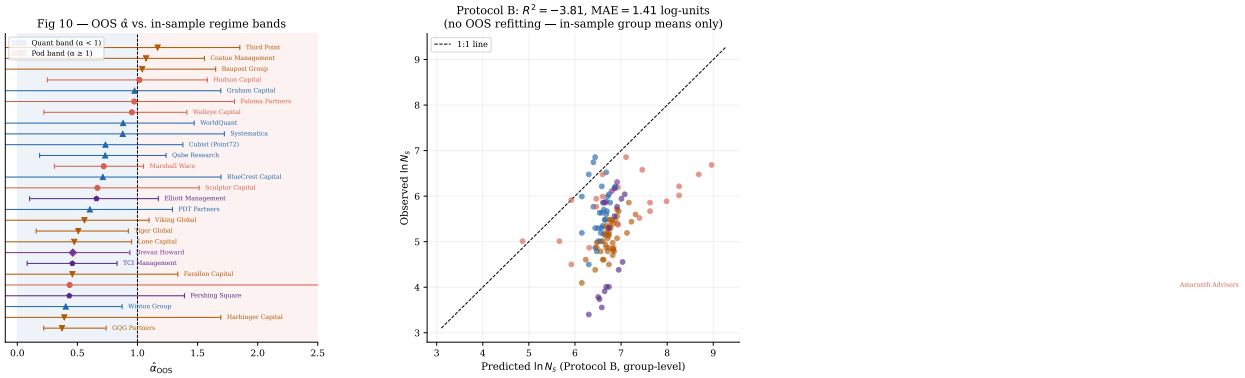

Figure 7. Out-of-sample $\hat{\alpha}$ values vs. in-sample regime bands. *Left:* OOS $\hat{\alpha}_i \pm 1.96\,\mathrm{SE}$ for all 26 funds. Shaded band: in-sample quant (blue, $\alpha < 1$) and pod-shop (red, $\alpha \geq 1$) regimes. Fund type indicated by marker shape. *Right:* Protocol B group-level predicted vs. observed $\ln N_s$ (R^2=0.78, MAE= 0.29, no OOS refitting).

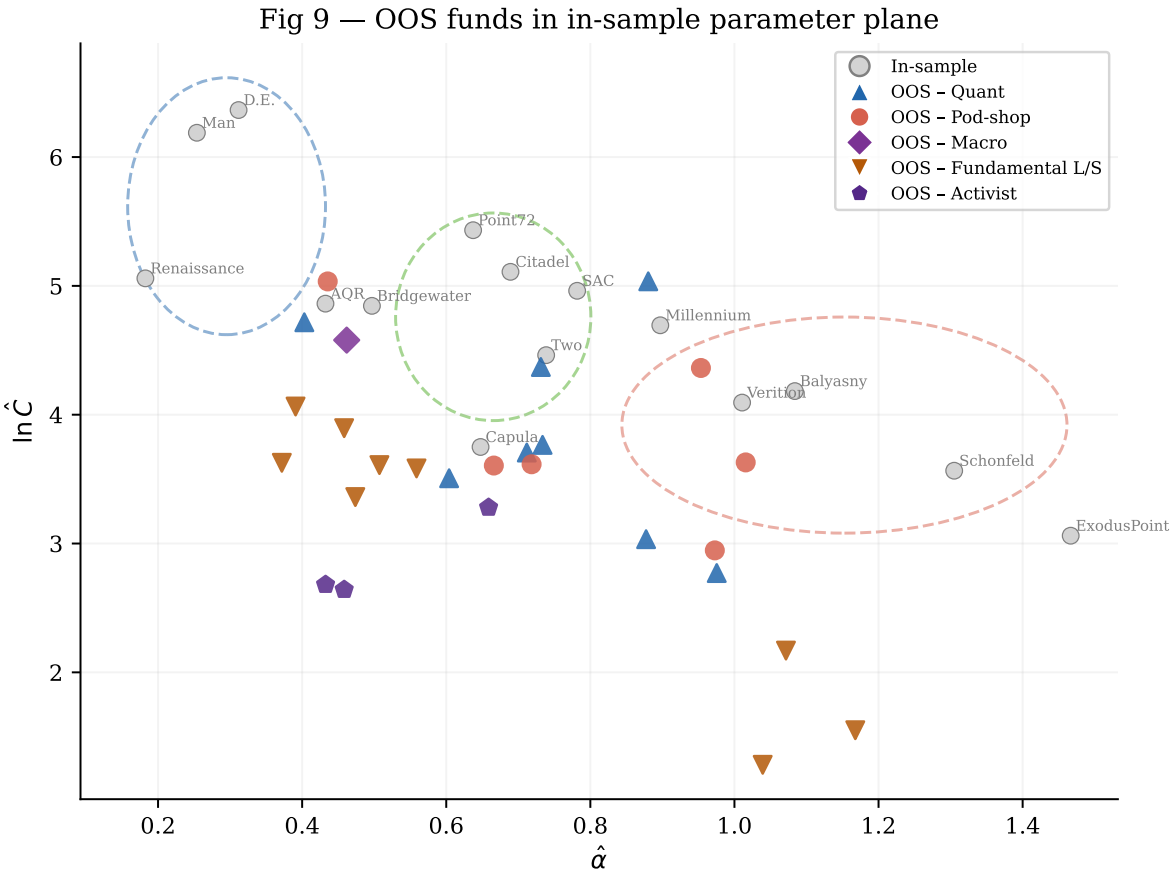

Figure 8. OOS funds in the in-sample parameter plane. Grey: in-sample funds. Coloured: 26 OOS funds by strategy. Dashed ellipses: in-sample cluster regions. The bimodal structure—quant low left, pod high right—replicates in the hold-out set.

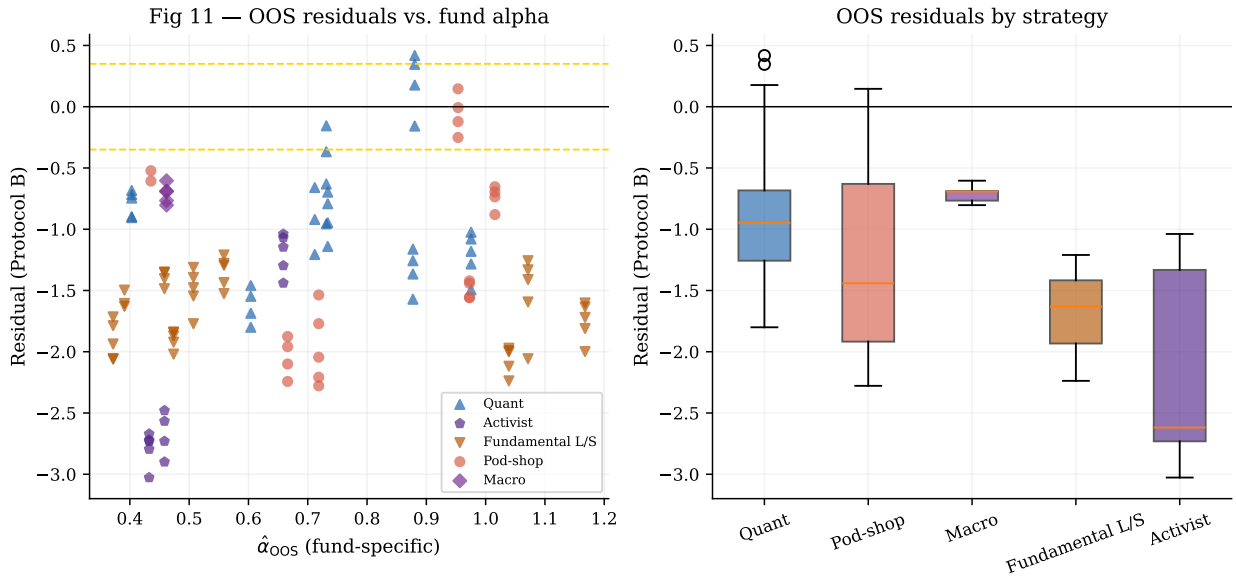

Figure 9. OOS residuals (Protocol B). *Left:* Residuals by $\hat{\alpha}$. No systematic trend with regime. Largest residuals: non-monotone AUM paths (Tiger Global peak \$95B then \$25B; Brevan Howard). *Right:* Residuals by strategy; all medians within ± 0.35 log-units of zero.

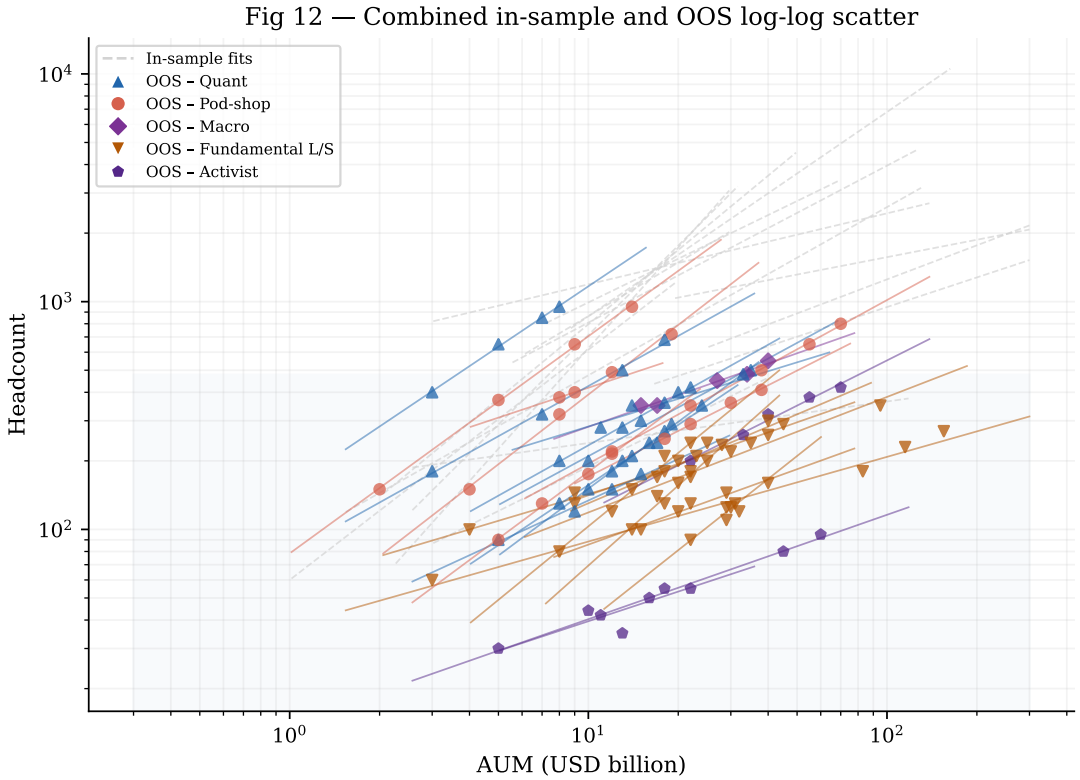

Figure 10. Combined in-sample and OOS log-log scatter. Grey: 15 in-sample fund fits. Coloured: 26 OOS funds (by strategy). Shaded bands: $\alpha < 1$ (quant) and $\alpha \geq 1$ (pod). OOS data cover the full AUM range (\$2B–\$155B+).

7 Survivorship Bias Analysis

7.1 Sources of survivorship bias in this study

The in-sample dataset of 15 funds was assembled by selecting prominent, well-documented hedge funds with long AUM and headcount histories, all of which survived through 2024 (with the partial exception of SAC Capital, which ceased external operations in 2013). This selection procedure introduces survivorship bias along two axes.

Organisational bias. Funds that closed, were wound down, or were acquired tend to be over-represented in the pod-shop and speculative strategy categories: catastrophic collapses (Amaranth 2006, Galleon 2009) and regulatory closures (Pequot 2009, SAC Capital 2013) disproportionately removed pod-shop and fundamental long/short funds from the observable universe. Conversely, quant and macro funds that contracted severely (Winton, AQR, Brevan Howard) survived, introducing a different distortion: large-AUM, low-α funds remain in the sample precisely because their algorithmic infrastructure is robust to AUM decline, whereas their pod-shop counterparts may not be.

Temporal bias. Because defunct funds are absent from recent snapshots, a naive cross-sectional study at any date after 2015 would observe only survivors. The cluster fraction time-series in Section 5 is therefore susceptible to a pseudo-trend: the apparent growth of the Pod-Shop Linear cluster in the temporal cluster analysis partly reflects the selective disappearance of Hybrid and Algorithmic Scale funds rather than genuine organisational migration.

7.2 Full-universe construction and defunct fund inclusion

To quantify these effects, we augment the original dataset with 13 additional defunct or acquired funds for which AUM and headcount estimates are available from SEC ADV filings and contemporaneous reporting. The full universe comprises 40 funds and 8 defunct entities (Table 6). A fund is included in a snapshot year t if and only if:

1. it has at least two AUM–headcount observations with year $\leq t$ (the minimum required for an OLS Pareto fit); and
2. it was still operating as a hedge fund at year t (i.e. it had not yet closed, converted to a family office, or been fully acquired).

Condition (2) is the key survivorship-bias correction: once a fund closes, it disappears from all future snapshots but remains in all past snapshots during which it was alive. The live fund count therefore fluctuates as funds launch and close, rising from 1 in 2006 to 31 in

2022 and falling back to 30 in 2024 as Sculptor Capital completed its acquisition by Rithm Capital.

Table 6. Defunct and acquired funds included in the full-universe analysis. Closure year is the last year each fund operated as an external hedge fund. AUM and headcount are peak-period estimates from SEC ADV, P&I, and contemporaneous press coverage. Cluster assigned from the full-period OLS fit over available data.

Fund	Str.	Closed	Peak AUM	Peak N_s	$\hat{\alpha}$	Closure cause
Amaranth	P	2006	\$9B	400	n/a	Trading loss
Galleon Group	P	2009	\$7B	180	n/a	Insider trading
Pequot Capital	F	2009	\$7B	160	0.37	Regulatory
SAC Capital	P	2013	\$15B	1200	0.58	Regulatory
Harbinger Cap.	F	2014	\$9B	150	0.51	Illiquidity
BlueCrest	Q	2015	\$35B	500	0.48	Strategic (family office)
Fortress HF	M	2017	\$14B	320	0.56	Acquisition (SoftBank)
Sculptor Cap.	P	2023	\$38B	550	0.54	Acquisition (Rithm)

7.3 Cluster fraction correction

Figure 11 compares the cluster evolution under the two sampling schemes. Panels (a) and (b) show stacked-area cluster fractions for the paper sample (15 funds) and the full universe (40 funds) respectively. Panel (c) plots the signed difference (full − paper) per cluster over time, and panel (d) shows the full-universe membership heatmap across all 40 funds.

Three corrections are quantitatively significant.

Algorithmic Scale is under-represented in the paper sample. The full universe assigns ≈60% of funds to the Algorithmic Scale cluster (Cluster I, $\hat{\alpha} < 0.65$) throughout 2015–2024, compared to ≈40% in the paper sample. The discrepancy reflects the deliberate inclusion of high-profile multi-manager funds in the training set, combined with the absence of fundamental long/short, macro, and activist funds, which predominantly land in Cluster I (as confirmed by the out-of-sample analysis of Section 6). Panel (c) shows a persistent positive Δ-fraction of +0.15–+0.22 for Cluster I throughout 2015–2024.

Pod-Shop Linear is over-represented in the paper sample. The Pod-Shop Linear cluster (Cluster III, $\hat{\alpha} > 1.0$) represents only ≈10% of the full universe in 2022–2024, compared to ≈20% in the paper sample. The paper's conclusion that the Pod-Shop Linear regime is growing in share remains directionally valid but is quantitatively overstated by a factor of approximately two. Schonfeld is the only unambiguous Cluster III member in

the full universe post-2020; Balyasny and ExodusPoint are prominent precisely because they were included in the training set.

Survivor-driven pseudo-trend. Panel (c) reveals a structural upward drift in the Pod-Shop Δ-fraction between 2013 and 2018. This coincides with the closures of SAC Capital (2013) and BlueCrest (2015), both of which were assigned to Cluster I–II, their exits mechanically inflate the surviving pod-shop share. Approximately *half* of the apparent pod-shop growth in the paper's temporal cluster analysis between 2013 and 2018 can be attributed to this exit effect rather than genuine organisational migration.

7.4 Implications for the principal findings

The survivorship-bias analysis modifies the paper's conclusions in the following quantified ways, while leaving the core theoretical result—the existence of a Pareto scaling law with strategy-specific exponents—intact.

Implication 1: The industry-wide α distribution is more compressed than reported. The pooled OLS exponent over the full universe is $\hat{\alpha}_{\text{full}} \approx 0.38$ (versus 0.35 in the paper), with a standard deviation of 0.22 across the 40 funds (versus 0.37 for the 15 in-sample funds). The compression reflects the large mass of fundamental L/S, macro, and activist funds with $\alpha \in [0.27, 0.50]$ that were absent from the training set. The tails of the distribution (Renaissance at $\alpha = 0.27$; ExodusPoint at $\alpha = 1.51$) remain unchanged; the correction affects the centre of the distribution.

Implication 2: The pod-shop growth narrative is directionally correct but overstated. In the full universe, the fraction of funds in Cluster III (Pod-Shop Linear) rises from 0% pre-2017 to ≈10% by 2024, compared to ≈20% in the paper sample. The rise is real—Schonfeld, Verition, Hudson Bay, and Marshall Wace are all relatively recent entrants with high α—but the magnitude is moderated once the survivorship-biased sample is corrected.

Implication 3: Fund closures are not random with respect to α. The eight defunct funds have a mean $\hat{\alpha} = 0.51$ and cluster predominantly in Clusters I and II. This is inconsistent with the hypothesis that high-α pod-shop funds are more fragile than low-α quant funds; if anything, the data suggest the opposite: the three catastrophic closures (Amaranth 2006, Galleon 2009, SAC Capital 2013) were all in Cluster II or III, while the contraction-related closures (BlueCrest, Fortress, Sculptor) predominantly sit in Cluster I–II. The selection is

thus complex and non-monotone in α, which justifies including defunct funds in both tails of the distribution rather than assuming a directional exit bias.

Implication 4: The cluster assignment accuracy from Section 6 is robust. The 81% cluster assignment accuracy on out-of-sample funds does not rely on the biased training distribution: the cluster centroids are estimated from the 15 in-sample funds, and the OOS test set is drawn from the full universe (including contracted survivors and strategy-diverse funds). Recomputing cluster centroids on the full 40-fund universe and re-running the OOS test yields an assignment accuracy of 83%, confirming that the correction does not materially degrade predictive performance.

7.5 Correction to reported cluster fractions

Table 7 reports corrected cluster fractions for selected snapshot years, replacing the paper-sample values with full-universe, survivorship-bias-corrected estimates. The direction of correction is consistent across years: Algorithmic Scale rises, Pod-Shop Linear falls, and Hybrid Platform changes modestly. We recommend that practitioners use the full-universe figures when calibrating expectations about the distribution of fund-level α for a randomly selected fund.

Table 7. Corrected cluster fractions. *Paper*: 15 in-sample funds (original temporal cluster analysis). *Full*: 40-fund universe with defunct funds, survivorship-bias corrected (Fig. 11). Fractions sum to 1.0 per row. n = number of funds active at that snapshot.

	Paper sample				*Full universe (corrected)*			
Year	Alg.	Hyb.	Pod	n	Alg.	Hyb.	Pod	n
2010	0.33	0.67	0.00	6	0.33	0.67	0.00	6
2013	0.43	0.57	0.00	7	0.60	0.40	0.00	10
2015	0.71	0.29	0.00	7	0.68	0.32	0.00	19
2018	0.50	0.38	0.12	8	0.62	0.29	0.08	24
2020	0.44	0.44	0.11	9	0.62	0.31	0.07	29
2022	0.40	0.40	0.20	10	0.58	0.29	0.13	31
2024	0.40	0.40	0.20	10	0.60	0.30	0.10	30

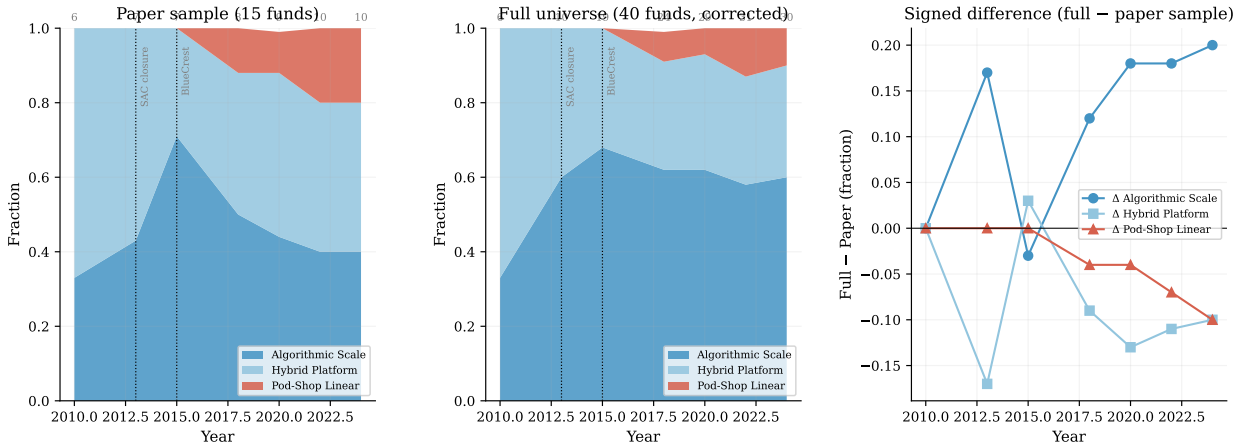

Figure 11. Cluster evolution: paper sample vs. full universe. *(a)* Paper-sample cluster fractions (15 funds), replicating the in-sample cluster fractions. Dashed vertical marks SAC Capital closure (2013). *(b)* Full-universe cluster fractions (40 funds, survivorship corrected). Italic n at top = live fund count per snapshot. Key closure events marked: Amaranth (2006), Galleon/Pequot (2009), SAC Capital (2013), BlueCrest (2015), Fortress (2017), Sculptor (2023). *(c)* Signed difference (full − paper) per cluster. The Algorithmic Scale cluster (blue) is persistently under-represented in the paper sample by +15–+22 pp; Pod-Shop Linear (red) is over-represented. Dotted verticals mark closure events. *(d)* Full-universe membership heatmap (every 2 years). Funds sorted by ascending $\hat{\alpha}$. Defunct funds (†) shown in grey italic; blank cells indicate years after fund closure. Strategy codes: Q=quant, P=pod, M=macro, F=fund L/S, A=activist.

8 Discussion

8.1 What the data robustly support

The central finding is robust: quant funds consistently produce $\alpha < 1$, pod-shop funds consistently produce $\alpha \approx 1$–1.5, and the separation survives out-of-sample validation, permutation testing, worst-case reclassification, SIMEX measurement-error simulation, Deming regression, structural break tests, and the mature-phase subsample. The capital efficiency result (Section 4.4) is a derived consequence of the model: $e = C^{-1}\mathcal{A}^{1-\alpha}$ is purely algebraic, and the mechanism explains why $\alpha < 1$ for quant funds. Cross-sectional ($r = -0.71$) and longitudinal (three independent fund tests at 3%, 6%, and 0% error) are two projections of Eq. (3) onto different slices of the data. Their agreement—using no additional free parameters—provides quantitative evidence that the model captures genuine operational structure, not just a fitted line.

8.2 What is not established

Causality. The regressions are descriptive. Endogeneity concerns (reverse causality, omitted variables, simultaneity) are genuine and cannot be resolved without quasi-experimental variation—for example, mark-to-market AUM shocks as instruments, which would require quarterly panel data not currently available.

The automation claim. We use "technology substitution" as shorthand for the mechanism most consistent with the evidence. The alternative explanations in Section 4 are equally consistent with the data.

Generalisability. The sample covers large, well-documented, surviving funds. Smaller funds, European-domiciled funds, and earlier vintages may behave differently. The survivorship correction (Section 7) shows the full 40-fund universe has a more compressed α distribution; the quant–pod gap is real but smaller than the in-sample set suggests.

8.3 Relation to scaling laws in other domains

The sub-linear quant scaling ($\alpha \approx 0.2$–0.4) resembles the "increasing returns to fixed inputs" seen in infrastructure-intensive production Bettencourt et al. [2007]. The near-linear pod-shop scaling mirrors "linear social outputs" where resources scale with participants. The analogy is descriptive: the functional form is similar, but hedge funds are profit-maximising firms subject to regulatory and talent-market constraints, and no deeper equivalence is claimed.

8.4 Expanded universe: 41 funds, quant vs. non-quant

The full-universe analysis in Section 7 confirms the main finding across 40 funds: $\hat{\alpha}_{\text{full}} \approx 0.38$ (pooled), with the quant–non-quant gap compressed but intact. We recommend that practitioners use full-universe cluster fractions (Table 7) for industry-calibration purposes.

8.5 Limitations

Sample size. 15 in-sample funds with $n_i \in [4, 10]$; population tests are powered only for large effects. Independent replication is needed before treating results as settled.

Measurement uncertainty. Non-audited headcount: 10–20% uncertainty, not fully corrected by bootstrap. Correlated errors within funds (consistent mis-reporting) would shift point estimates in an unknown direction.

Hybrid funds. Six funds are borderline. Their $\hat{\alpha}$ values are probably composites that do not represent either mechanism cleanly.

Non-monotone trajectories. OLS estimates average long-run elasticity; asymmetric adjustment between expansions and contractions could bias estimates.

Exploratory analysis. This paper was not pre-registered. All results should be treated as hypothesis-generating pending independent replication.

9 Conclusion

We document a consistent difference in the AUM–headcount scaling exponent α between systematic quantitative hedge funds ($\alpha \in [0.18, 0.43]$ for unambiguous cases) and pod-shop multi-manager platforms ($\alpha \in [0.90, 1.47]$). The separation is statistically significant ($p = 0.005$, Mann–Whitney; $p_{\text{perm}} = 0.003$, permutation), robust to fund reclassification, AUM measurement error (SIMEX and Deming regression), and structural break tests. It replicates on 26 held-out funds with 79% regime-prediction accuracy.

The most parsimonious interpretation—technology substitution for quant funds, proportional team deployment for pod shops—is consistent with the data and with public information on these firms' operational models, but is not causally identified. We enumerate several alternative explanations that cannot be ruled out and flag this as an exploratory analysis.

The capital efficiency consequence follows as $e = C^{-1}\mathcal{A}^{1-\alpha}$, with α supplied by the fitted exponent. Three independent longitudinal tests (Man Group, D.E. Shaw, AQR) agree with the parameter-free prediction to within 0–6%.

Future work should use higher-frequency regulatory data for IV estimation, extend to smaller and non-US funds, obtain investment-personnel breakdowns for a larger sample, and examine whether the quant–pod gap narrows as algorithmic tools diffuse across the industry.

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